SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

Figure placeholder: Fig. 1 teaser — a role-aware 4D view of a moving scene (real robot)

Layout: full width, two parts. **(a) Left, ∼62%:** one real-robot frame of the *linear-motion grasp* at the instant a chunk is issued. Overlays in the paper's semantic colors: blue = the target's past SE(3) poses from the causal tracker (small triads at $-350 \ldots 0$ ms, fading with age); orange = the target's SE(3) forecast at the execution times of the chunk, taken from the training-time heads evaluated offline; light orange arrows = sparse 3D flow on the context on the 14×14 grid only; purple = the commanded gripper path meeting the target at τ^{a}. A gray ghost marks the pose observed at capture, with a dashed red arrow showing where a reactive policy would close. A thin timeline strip under the image: capture $\tau_t \rightarrow$ latency $\ell_t \rightarrow$ execution τ^{a}. **(b) Right, ∼38%:** top: "Image-space WAM: future frames or latents at uniform fidelity" (filmstrip icon); middle: "SE(3)-WAM: SE(3) for the target, sparse 3D flow for the context, training only" (triads + sparse arrows); bottom **(required)**: SR vs. added latency (0–200 ms) for SE(3)-WAM and the strongest baseline (data: the latency sweep of Appendix D.2). **Rules:** real photo kept unretouched; overlays drawn from logged tracker/forecast/command data of that trial; caption states the trial's outcome.

Figure 1. **A role-aware 4D view of a moving scene.** Between image capture and the first command, a part on the conveyor moves farther than the grasp tolerates. SE(3)-WAM encodes the target's metric motion (blue) and is trained to forecast the target's SE(3) pose precisely and the whole scene coarsely as a sparse 3D flow (orange), so that its commands meet the target when they execute (purple). Image-space world–action models forecast appearance at uniform fidelity instead. Right: success against added latency on Dynamic-14.

Abstract

Manipulating moving objects requires actions that fit the scene when they execute, not when the image was captured, yet every millisecond spent inferring the scene's motion lets the target move further. World–action models forecast future images or latents at uniform fidelity, spending time on appearance while leaving the target's metric pose implicit; target-centric predictors are fast but ignore the surroundings that constrain the motion. We observe that the parts of a dynamic scene need different precision: the target's pose decides contact to within millimeters, whereas the rest of the scene matters only through the clearance it leaves. We present SE(3)-WAM, a world–action model built on this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history and is trained to forecast the target's future SE(3) trajectory precisely and the whole scene coarsely as a sparse 3D flow. The forecasts serve only as training supervision and are skipped at deployment, so inference needs neither past frames nor a generated future and stays fast enough for real-time control. On Dynamic-14, a simulated benchmark in which the scene keeps moving during inference, SE(3)-WAM achieves [XX.X]% success, [XX.X] points above the strongest baseline, and succeeds in [XX]% of trials on six real-robot tasks with diverse target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [9, 16, 37, 56, 75, 84, 106], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. This poses two coupled difficulties. A single image does not reveal how the target is moving, and by the time an action executes, the target is no longer where it was seen: a part on a conveyor at 0.1 m/s travels 15 mm during a 150 ms delay, more than a parallel-jaw grasp typically tolerates (Fig. 1).

Recent work equips policies with a memory of past motion and a forecast of the future [19, 49, 67, 94]. Both cost the resource that dynamic manipulation lacks most, time: richer histories and denser forecasts add computation, and the added latency enlarges the very error the forecast is meant to remove. Existing designs therefore trade information against speed. Video-based WAMs forecast whole future frames or latents at uniform fidelity [49, 75, 84], and methods that restrict prediction to dynamic regions or salient tokens still predict in image space [67, 94]. In both, most capacity goes to appearance, and the quantity that decides contact, the target's metric pose when the action executes, stays implicit. Target-centric predictors sit at the other end [19, 28, 50]: they are fast but blind to surroundings that

can block or redirect the motion. None of them provides knowledge of the moving scene that is precise where contact depends on it, coarse elsewhere, and cheap to use at execution time.

Our key observation is that a dynamic policy needs to know very little precisely. Contact depends on the target's pose at the moment the action executes, to within millimeters, whereas the rest of the scene matters only through the clearance it leaves. Nor does this knowledge have to be computed at test time: a forecast that supervises training shapes what the policy represents and can be dropped at deployment [89]. SE(3)-WAM builds on both ideas (Fig. 2). It sees the current image together with short metric SE(3) histories of the target and the robot, which convey motion directly instead of a stack of past frames. A world stream is trained to forecast, in one metric 4D space, the target's future SE(3) trajectory precisely and the whole scene coarsely as a sparse 3D flow, and an action stream reads the world stream's summary and replans as often as its latency permits. SE(3)-WAM thus gains the precision of a metric forecast at a latency close to that of a reactive policy.

We evaluate SE(3)-WAM on Dynamic-14, a simulated benchmark of fourteen tasks in which observation, scene motion, and action execution share one physical clock, so the scene keeps moving while a policy computes. SE(3)-WAM achieves state-of-the-art success of [XX.X]%, [XX.X] points above the strongest baseline, at a p95 latency of [XX] ms, and succeeds in [XX]% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Four final-scale controls isolate the contributions of target-motion history, future supervision, and its two geometric components.

Our contributions are threefold:

- We propose role-aware 4D modeling for dynamic manipulation. Tracing how latency and estimation errors reach the contact shows that only the target needs millimeter precision, so we represent the target precisely by its metric SE(3) pose history and future trajectory, and the whole scene coarsely by a sparse 3D flow.
- We present SE(3)-WAM, a world–action model trained with these role-aware 4D forecasts as supervision, at a latency close to that of a reactive policy.
- We build Dynamic-14, in which the scene keeps moving during inference, and show state-of-the-art results on it and on six real-robot tasks.

2. Related Work

Dynamic manipulation. Compensating for the time between sensing and acting is a classical problem, from control with dead time [64] to catching systems that track the target with Kalman-type filters and plan an interception [2, 5, 6, 52]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes. PUMA reads historical optical flow and is supervised with future object features [19], DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [49], and other methods forecast salient tokens, future images, or compact future codes and feed them to the policy at test time [50, 67, 90, 95]. Execution schemes instead reduce or hide latency, by overlapping inference with execution [60, 62, 81], keeping consecutive chunks consistent [10, 46], or conditioning on the execution time [68, 72]. They decide when actions are computed and applied, whereas SE(3)-WAM changes what the policy represents; we run every method under the same real-time executor.

Predictive supervision in world–action models. World–action models (WAMs) generate future video and actions in one network [40, 75, 84, 103], building on video prediction as a policy objective [29, 70, 79]. Because video is costly and dominated by appearance, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [94], flow and geometry alongside RGB [88], object masks or slots [48, 85], or compact latents [38, 39], and asks whether the forecast is needed at test time: Fast-WAM retains most of the benefit of video modeling without test-time imagination [89], and PUMA and others drop their predictors after training [19, 66, 97]. SE(3)-WAM adopts this training-only use and changes the target of the supervision: instead of pixels, features, or latents at one resolution, a metric 4D state whose precision follows each entity's role in contact.

Geometric representations of motion. Metric 3D inputs make policies more precise and sample-efficient [22, 35, 61, 91]. Motion has been represented by 2D point tracks [7, 77, 82, 100], 3D flow [23, 30, 87, 101], and object keypoints or SE(3) trajectories [28, 51, 65, 78], and rigid-body decompositions have long been used to learn scene dynamics [12, 13]. These works describe the scene with one kind of representation, dense flow or tracks everywhere or a pose for each object; DOCIR separates the robot, the target, and obstacles, but as input encodings [18]. SE(3)-WAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, and the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast. Appendix F discusses further related work.

3. SE(3)-WAM

SE(3)-WAM is a world–action model that represents a dynamic scene at role-dependent precision. As shown in Fig. 2, the current image and short metric SE(3) histories of the target and the robot form a world stream, which is trained to forecast the target's SE(3) trajectory and a sparse 3D flow

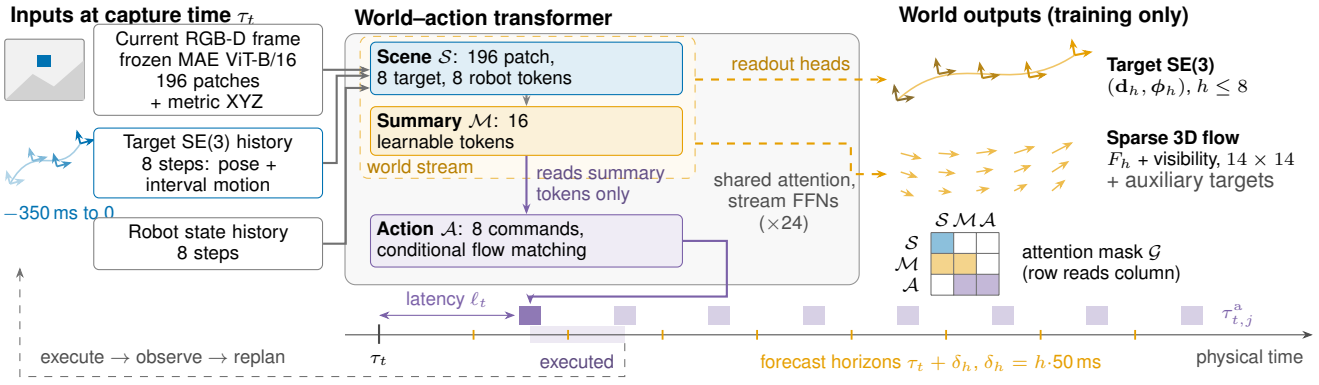


Figure 2. **SE(3)-WAM.** The world stream holds 212 scene tokens, the current image patches from a frozen encoder together with short histories of the target's metric SE(3) pose and of the robot state, and 16 summary tokens. During training, light heads read the world stream out into a role-aware 4D forecast at physical horizons $\tau_t + \delta_h$: the target's SE(3) trajectory and a sparse 3D flow with visibility on a 14×14 grid over the whole scene (dashed: training only). The action stream generates a chunk of 8 commands by flow matching. The attention mask $\mathcal{G}$ lets scene tokens read the scene, summary tokens read the scene and summaries, and actions read summaries and actions. The policy observes and replans at the first control boundary at or after each query ends.

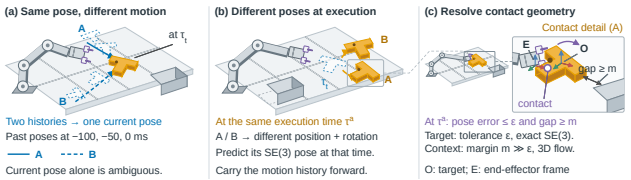


Figure 3. **Acting on a moving target** (schematic). (a) Two histories reach the same pose at capture time τ_t but imply different motions. (b) At the execution time τ^{a}, they lead to different target poses. (c) A contact requires the end-effector frame E to match the target O within ε, whereas the context matters only through the gap it leaves, $m \gg \varepsilon$.

of the whole scene. An action stream shares attention with the world stream and generates action chunks by conditional flow matching [43].

3.1. Acting at Execution Time

At query t, a policy receives an image, a depth map, and the robot state captured at time τ_t, together with histories of earlier states, and outputs a chunk $(\mathbf{a}_{t,0}, \ldots, \mathbf{a}_{t,H-1})$ whose j-th command executes at

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + \ell_t + j\Delta_a, \tag{1}$$

where ℓ_t is the latency from capture to the first execution and Δ_a is the control period. A command must fit the scene at $\tau^{\mathrm{a}}_{t,j}$ rather than at τ_t (Fig. 3), and the lead $\Delta = \tau^{\mathrm{a}}_{t,j} - \tau_t$ grows with every millisecond of inference.

Where the error comes from. Let a policy act, explicitly or implicitly, on an estimate of the target pose at τ^{a} with position error e and rotation error θ. A contact point at lever arm r is then off by at most $e + 2r\sin(\theta/2)$. Writing the target's motion over the lead as a constant velocity $\mathbf{v}$ plus a residual (acceleration, rolling, impacts), and the lead that the policy assumes as $\hat{\Delta}$, gives

$$e \le e_{\mathrm{loc}} + \hat{\Delta}\, e_v + \|\mathbf{v}\|\, e_\tau + e_{\mathrm{dyn}}, \tag{2}$$

with the errors e_{loc} of the current position, e_v of the velocity, $e_\tau = |\hat{\Delta} - \Delta|$ of the timing, and e_{dyn} of the residual motion (Appendix E). Two observations guide our design. First, the precision that matters sits on the target: its pose and motion must be known to within millimeters, and its rotation error is amplified by the lever arm, whereas the surroundings matter through the clearance m they leave, which is usually far larger than the contact tolerance ε. Second, latency enters twice, through the lead that multiplies every velocity error and through the motion that the residual accumulates, so information that costs inference time pays off only if it removes more error than the added latency creates.

3.2. Model

Inputs. SE(3)-WAM sees one image. A frozen MAE ViT-B/16 [26] encodes the current RGB frame into 196 patch tokens, and the depth map gives each patch the metric 3D position of its center point in the robot base frame. Motion comes from two short histories over the last 8 control steps (350 ms at 20 Hz). The target history consists of the current target pose in the base frame and the seven interval motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units (Appendix A.1). The robot history consists of 8 end-effector states. Together with the patch tokens, they form 212 scene tokens. In simulation, the target pose comes from the simulator; on the real robot,

a causal tracker estimates it from the RGB-D stream (Appendix C). The model receives no past images, so the cost of visual encoding does not grow with the history.

World and action streams. The world stream holds the scene tokens and 16 learnable summary tokens. The action stream holds 8 tokens, one per command, each embedding a noisy command; a chunk spans 8 control periods of $\Delta_a = 50$ ms, and each command is an end-effector translation and rotation plus a gripper command. The attention mask $\mathcal{G}$ lets scene tokens attend to the scene, summary tokens to the scene and the summaries, and action tokens to the summaries and the actions. In every layer, both streams share the attention projections and have separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time.

3.3. Role-Aware 4D Supervision

The forecast heads supervise the world stream at fixed physical times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50$ ms and $h = 1, \dots, 8$, from 50 ms to 400 ms after capture. All motion labels are cumulative from the current state, and every loss averages over valid labels only.

Target SE(3) trajectory. For each horizon, the target's motion is

$$\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t), \quad \boldsymbol{\phi}_h = \mathrm{Log}\big(R(\tau_t{+}\delta_h)R(\tau_t)^\top\big)^\vee, \tag{3}$$

where $(R, \mathbf{p})$ is its pose in the base frame. A linear pose head on the mean of the final summary tokens predicts $(\hat{\mathbf{d}}_h, \hat{\boldsymbol{\phi}}_h)$, and

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{4}$$

where $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^\wedge) R(\tau_t)$ and $\mathcal{V}$ is the set of valid horizons.

Sparse 3D flow of the whole scene. The center points of the 196 patches on the 14×14 grid cover the target, the robot, and the surroundings. For each point i and horizon h, a linear head on its final patch token predicts the base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, and

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h) \in \mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{5}$$

with a fixed scale s fitted on the training labels, the visibility o_{ih}, and the set $\mathcal{V}_F$ of valid labels. The grid is coarse, about [X] mm between points in the workspace. The flow thus gives every part of the scene a coarse motion estimate, and the target additionally receives the precise SE(3) supervision of Eq. (4); this is how SE(3)-WAM allocates precision by role.

Auxiliary targets. Two further targets regularize the representation. Future appearance heads on the patch tokens predict, at every horizon, the metric depth, the features of a frozen DINOv3 teacher [63], and the RGB of moving surfaces. A small head on the summary tokens reconstructs the RGB of the current target region, which ties the summaries to the target without any future label. Their losses form $\mathcal{L}_{\mathrm{aux}}$ (Appendix A.2).

3.4. Training and Inference

Training. For a normalized chunk $\mathbf{a}_1$, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream regresses the velocity

$$\mathcal{L}_A = \mathbb{E} \left\| v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t, \mathcal{G}) - (\mathbf{a}_1 - \mathbf{a}_0) \right\|^2, \tag{6}$$

with $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, the scene tokens $\mathcal{S}_t$ of query t, and the error averaged over valid commands. All parts are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_P \mathcal{L}_P + \lambda_F \mathcal{L}_F + \lambda_{\mathrm{aux}} \mathcal{L}_{\mathrm{aux}}. \tag{7}$$

Inference. As in recent WAMs [40, 89], the forecast heads are trained jointly and skipped at test time. Because world tokens never attend to actions, the world stream runs once per query, and the action stream integrates 5 Euler steps on its cached keys and values. A new chunk starts at the first control-period boundary after its query ends, where the policy observes again and starts the next query; until then, the executor continues the previous chunk. The policy thus replans as often as its latency allows, before every command when a query fits in one control period (Appendix A.6).

4. Experiments

We compare SE(3)-WAM with reactive, dynamic, and world–action policies in simulation (Sec. 4.2) and on a real robot (Sec. 4.3), and then ablate its target history and its geometric supervision (Sec. 4.4).

4.1. Setup

Dynamic-14. We built fourteen tasks in MuJoCo/robosuite [71, 104] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Appendix B, Fig. S2). Experts with privileged simulator state provide 800 demonstrations per

task. We select configurations on development scenes and report all Dynamic-14 results on 1,400 held-out scenes (100 per task, new seeds). Our variants average three training seeds, and each baseline is trained once. We give paired 95% intervals from a bootstrap over held-out scenes.

Protocol. Observation, scene motion, and execution share one physical clock with a control period of 50 ms, and physics advances during inference by a latency drawn from each method's measured latency on one RTX 5090. All methods run under the same executor: a query starts at a control-period boundary, its chunk replaces the remainder of the previous chunk at the first boundary after the query ends, and the next query starts there (Appendix B.3). The +150 ms setting adds 150 ms of delay for every method, and the paused setting stops physics during inference. We report the task-balanced success rate (SR), the position error at first contact e_p, and the p95 time-to-action.

Baselines. We compare with reactive policies (ACT [98], Diffusion Policy [16], and $\pi_{0.5}$ [56]), dynamic VLAs (DynamicVLA [81] and PUMA [19]), image- or latent-space WAMs (DreamVLA [94] and OpenWAM-α [75]), and a dynamic WAM (DynamicWAM [49]). Each baseline uses its official code, inputs, and recipe, is trained on the same demonstrations, and receives a rendered wrist camera if its recipe uses one (Appendix B.4). SE(3)-WAM uses only the agent view, together with depth and the target pose. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)). The *CV input* variant is SE(3)-WAM without forecast losses that instead receives the constant-velocity extrapolation of the target pose as input tokens.

Implementation. The transformer has 24 layers of width 1024 with 16 heads and 523.8M trainable parameters; the MAE ViT-B/16 encoder and the DINOv3 teacher stay frozen. On Dynamic-14, we train from a randomly initialized transformer on all windows of the demonstrations for 20 epochs (51,740 updates at a global batch size of 512) with AdamW, a cosine schedule, $\lambda_P = \lambda_F = 1$, and $\lambda_R = \lambda_{\text{aux}} = 0.1$, which takes [X] hours per seed on [N] RTX PRO 6000 GPUs (Appendix A.4). For the real robot, we pretrain a 48-layer base (1.02B parameters) on DROID [36], whose demonstrations carry no target poses: it learns the actions, the sparse 3D flow from the point tracks that PointWorld released [30], and the future DINOv3 features, for 21,060 updates on 64 GPUs. Fine-tuning on each task adds the target-state tokens and the pose head (Appendix A.5).

4.2. Simulation Results

SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name of the strongest baseline], and [X.X] points above the strongest baseline that receives the same SE(3) history, [name]. The margin is largest on [rotating targets and on fast conveyor and translating targets], where the target moves farthest during the lead, and its position error at first contact is [X.X] mm against [X.X] mm for [name]. Its p95 time-to-action of [XX] ms is [X]× lower than that of the image-space WAMs, whose latency adds to the lead they must bridge. With 150 ms of added delay, SE(3)-WAM keeps [XX.X]%, whereas reactive policies drop to [XX.X]% and image-space WAMs to [XX.X]%. With physics paused during inference, the difference to [name of the strongest paused baseline] is [±X.X] points, which separates the contribution of speed from that of the representation. The CV-input variant, which extrapolates the target's current motion instead of learning a forecast, trails SE(3)-WAM by [X.X] points.

DOMINO. On a second benchmark, we compare SE(3)-WAM with $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [19], with its demonstrations and native evaluator. SE(3)-WAM reaches [XX.X]% SR against [XX.X]% for [the strongest baseline] (Appendix D.3). As on Dynamic-14, it reads the simulator pose of the target.

4.3. Real-Robot Experiments

We evaluate six tasks on a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera (Fig. S3, Table S8), comparing SE(3)-WAM with $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [16]: conveyor grasping (T1), placement into a receptacle on the conveyor (T2), turntable grasping (T3), grasping a moving toy car (T4), intercepting a ball after a collision (T5), and intercepting an object that rolls or slides down a ramp (T6). Together they cover linear, circular, free planar, post-impact, and gravity-accelerated target motion. Each method is trained or fine-tuned per task on the same 100 teleoperated demonstrations, with 20 separate validation demonstrations, and runs 20 trials per task with shared trial keys. SE(3)-WAM tracks the target with [tracker] and runs on one RTX 5090 with a p95 time-to-action of [XX] ms. It succeeds in [XX.X]% of all trials, against [XX.X]% for [the strongest baseline] (Table 2). [Where its advantage is largest, and its dominant failure mode] (Fig. 4).

4.4. Ablations

Four controls each change one component of the final model and keep the data, recipe, and training budget, with three training seeds each (Table 3, Fig. 5). We evaluate them under the real-time protocol at 0 and 150 ms of added delay.

Target history. The no-history control keeps the current image, the current target pose, and the robot history, and

Table 1. **Main results.** Held-out SR (%) on Dynamic-14 (1,400 new scenes, task-balanced, real time, three training seeds for our variants and one for baselines) by family, with 150 ms of added delay, and with physics paused during inference, together with the position error at first contact e_p (mm) and the p95 time-to-action on one RTX 5090. Inputs: R, RGB and robot state. D, depth. P, target SE(3) history. Best in bold. Intervals in Appendix D.

		Dynamic-14 SR, real time unless noted									
Method	Inputs	All	Grasp	Place	Pass	Capture	Impact	+150 ms	Paused	e_p	p95 (ms)
ACT [98]	R	—	—	—	—	—	—	—	—	—	—
Diffusion Policy [16]	R	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ [56]	R	—	—	—	—	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—
DynamicVLA [81]	R	—	—	—	—	—	—	—	—	—	—
PUMA [19]	R	—	—	—	—	—	—	—	—	—	—
DreamVLA [94]	R	—	—	—	—	—	—	—	—	—	—
OpenWAM-α [75]	R	—	—	—	—	—	—	—	—	—	—
DynamicWAM [49]	R	—	—	—	—	—	—	—	—	—	—
DynamicWAM + SE(3)	R+P	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM, CV input	RD+P	—	—	—	—	—	—	—	—	—	—
SE(3)-WAM	RD+P	—	—	—	—	—	—	—	—	—	—

Table 2. **Real-robot results.** Success rate (%) on six tasks: T1 conveyor grasping, T2 moving-receptacle placement, T3 turntable grasping, T4 moving-car grasping, T5 post-collision interception, and T6 ramp interception (20 trials per task and method). All pools the trials; intervals in Appendix C.

Method	T1	T2	T3	T4	T5	T6	All
$\pi_{0.5}$	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—

Figure placeholder: Fig. 4 — real-robot trials

Rows: a success of SE(3)-WAM; the strongest baseline on the same trial key; a failure of SE(3)-WAM. **Columns:** the query frame, the first execution time τ^a, and contact. **Overlays:** blue tracked past poses; purple commanded gripper path. Real photos, unretouched; overlays from logged data of each trial.

Figure 4. **Real-robot trials** of SE(3)-WAM and [the strongest baseline] on the same trial key (top two rows) and a failure of SE(3)-WAM (bottom). [One-sentence description of the selected trials.]

removes the past target poses with their velocity and acceleration features. Without them, SR changes by [signed X.X] points at 0 ms and [signed X.X] points at 150 ms of added delay (Fig. 5a). Under synthetic pose noise, SE(3)-WAM keeps [XX.X]% SR (Appendix D.4).

Table 3. **Core ablations** of the final 24-layer model on held-out Dynamic-14 scenes (three training seeds). Each row changes one component of SE(3)-WAM (*). SR, SR_{150}: success (%) without and with 150 ms of added delay. e_p: position error at first contact (mm). p95: time-to-action (ms).

Variant	SR	SR_{150}	e_p	p95
SE(3)-WAM *	—	—	—	—
No target history	—	—	—	—
No future supervision	—	—	—	—
No scene-flow supervision	—	—	—	—
No dedicated SE(3)	—	—	—	—

Future supervision. The no-future control removes all forecast losses, the SE(3), scene-flow, and future appearance targets, and keeps the current-target RGB reconstruction. It changes SR by [signed X.X] points at 0 ms and [signed X.X] points at 150 ms, which measures the effect of future supervision as a whole.

The two geometric components. The other two controls keep all auxiliary targets. Removing the scene-flow supervision changes SR by [signed X.X] points; since the flow covers the whole scene, this measures coarse scene motion as a whole rather than the surroundings alone. Removing the dedicated SE(3) supervision leaves the target with only the coarse flow that every part of the scene receives and changes SR by [signed X.X] points, most on [task families]. Fig. 6 shows what the SE(3) head predicts at the execution time.

Figure placeholder: Fig. 5 — ablations (two panels)

(a) By task family. SR difference to the full model of each of the four controls, per task family, at 0 and 150 ms added delay. **(b) Success and latency.** p95 time-to-action vs. SR for the full model and the four controls, at 0 and 150 ms added delay. All panels reuse the same held-out scenes and three training seeds; show paired 95% bootstrap intervals.

Figure 5. **Ablations.** (a) Effect of each control per task family. (b) Success and latency of the full model and the controls under the real-time protocol. [Summarize the measured effects.]

Figure placeholder: Fig. 6 — forecasts at the execution time

Rows: a rolling capture (median forecast error) and a post-impact grasp. **Columns:** the query frame at τ_t, the frame at τ^{a}, and a 3D inset in the base frame. **Overlays:** blue past poses, orange forecast triads and coarse flow arrows from the training-time heads, gray ground truth at the same times, purple commanded path. Full set in Fig. S5.

Figure 6. **Forecasts at the execution time** on held-out Dynamic-14 episodes, read from the training-time heads. Top: the SE(3) forecast follows a rolling target that constant-velocity extrapolation overshoots. Bottom: before an impact, the regressed forecast lies between the possible outcomes.

5. Conclusion

We presented SE(3)-WAM, a world–action model for manipulating moving objects that spends precision where contact depends on it and time only where it pays off. It remembers the target's metric SE(3) motion instead of past frames, learns to forecast the target's SE(3) trajectory precisely and the whole scene coarsely as a sparse 3D flow, and lets its actions read a representation shaped by these forecasts without generating them at test time. On Dynamic-14, where the scene keeps moving while a policy computes, and on six real-robot tasks, it improves on reactive, dynamic, and world–action policies. Four final-scale controls examine the metric motion history, future supervision, and its two geometric components. [State the supported component-level conclusions after the core ablations.]

Limitations. SE(3)-WAM assumes a rigid target that moves on its own until contact, so deformable or articulated targets and post-contact dynamics are outside its scope. It depends on a pose estimate of the target, and tracker failures propagate to the policy; in simulation we read the pose from the simulator. The regressed target forecast cannot represent multimodal futures, e.g., before an impact. Our real-robot base learns from DROID without target poses, so its pose-related parts are learned only during fine-tuning. Finally, our tasks have a single target and slow context motion.

LLM Usage Statement

[To be completed before submission: actual tools used, affected parts of the research and manuscript, and extent of use.]

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Supplementary Material

This supplement gives the model and training details of SE(3)-WAM, including pretraining on real robot data (Appendix A), the benchmark and the evaluation protocol (Appendix B), the real-robot setup (Appendix C), additional results (Appendix D), the error decomposition that motivates the design (Appendix E), and additional related work (Appendix F). Colored dashes and bracketed entries denote values to be filled from the final runs.

A. Model and Training

A.1. Scene and Summary Tokens

Current image. A frozen MAE ViT-B/16 [26] encodes the 224×224 agent-view RGB into 196 patch tokens; we chose it over frozen DINOv3 encoders [63] and a CNN trained with the policy on development scenes (Appendix D.5). Each patch receives a view embedding, a linear embedding of its mean valid metric depth, and a linear embedding of the base-frame position of its back-projected center pixel, computed with the camera intrinsics and extrinsics. These positions also enter 3D rotary position embeddings of the patch queries and keys, and the same center points are the source points of the sparse 3D flow (Appendix A.2). Target, robot, summary, and action tokens receive no rotary embedding. Past images never enter the model, so the scene holds 212 tokens.

Target-motion history. The target history covers the last 8 control steps, at the offsets $-350, -300, \ldots, 0$ ms. Each of the 8 target-state tokens is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$ with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$. Slots 0–6 describe the seven intervals between consecutive offsets, and slot 7 holds the current absolute pose in the base frame. Velocities divide interval displacements by the elapsed time, and accelerations divide adjacent velocity differences by the spacing of the interval centers. We standardize each dimension with training statistics, zero invalid dimensions, and append three validity flags (pose, interval, and acceleration) before a $28 \to 1024 \to 1024$ GELU MLP with an additive relative-time embedding. A slot without any valid feature is masked out of attention. On the real robot, we resample the tracker's estimates (Appendix C.2) at the same offsets, linearly in translation and along the geodesic in rotation, without extrapolation and without interpolating across gaps longer than two camera periods.

Robot state and summary tokens. 8 end-effector states (position, rotation vector, and measured gripper opening fraction) at the same offsets form the robot tokens. 16 learned embeddings form the summary tokens. The policy has no language token, and the view, time, depth, and position embeddings modify existing tokens rather than adding slots.

A.2. Forecast Heads

All forecast heads are light readouts of the final hidden states of the world stream. They are trained jointly with the action stream and skipped at deployment, and no token reads their outputs.

Target. A linear map from the mean final state of the summary tokens gives $\{(\hat{\mathbf{d}}_h, \hat{\boldsymbol{\phi}}_h)\}_{h=1}^{8}$, trained with $\mathcal{L}_P$ of Eq. 4 and $\lambda_R = 0.1$. We use the left-relative rotation vector and the cumulative translation instead of a twist exponential, which would couple translation to rotation through its Jacobian. Squared-error training gives a point estimate, which averages over multiple possible futures, e.g., before an impact.

Sparse 3D flow. The flow is the cumulative base-frame displacement of the 196 patch center points at the 8 horizons, together with their visibility. A linear readout on each patch token gives its point's displacement and visibility logit at every horizon, trained with $\mathcal{L}_F$ of Eq. 5 and the scale $s = 0.7154605$ m. The visibility loss has weight 1 in simulation, and points that are valid but invisible still supervise their displacement. The points cover the target, the robot, and the surroundings alike.

Auxiliary targets. Linear readouts of the final patch tokens predict three future appearance targets at every horizon: metric depth and moving-surface RGB through 16×16 pixel readouts per patch, and the 1024-dimensional features of a frozen DINOv3 ViT-L/16 teacher [63] on the 14×14 grid. Depth uses squared error on valid pixels, the DINO features squared error after channel-wise layer normalization, and RGB squared error on surfaces that move by more than 1 mm, including the robot. A $1024 \to 256 \to 3072$ MLP with GELU and a final sigmoid reads the mean of the final summary tokens and reconstructs the current target's 32×32 RGB region with an absolute error. The four losses form $\mathcal{L}_{\text{aux}}$, each with weight $\lambda_{\text{aux}} = 0.1$.

A.3. Action Stream and Backbone

Commands. Each chunk has 8 seven-dimensional commands at a control period of 50 ms, each with a base-frame translation increment, a left-relative rotation vector, and a gripper command. Translation and nonzero rotation increments refer to the measured end-effector pose at application. Normalization uses the mean and population standard deviation of the valid issued commands of the training split. Flow matching uses a unit Gaussian in these coordinates with one flow time u per chunk, which the action tokens receive through adaptive layer normalization. Each prediction

position draws 8 independent noise–time pairs and averages their losses, which reduces the variance of the action gradient without enlarging the batch. At deployment, we integrate 5 Euler steps, reverse the normalization, decode the rotation vectors, and clip each axis to 0.05 m and 0.5 rad; the gripper follows the sign of its command.

Backbone and mask. Each of the 24 layers applies one attention over all tokens with shared query, key, value, and output projections, followed by separate SiLU feed-forward networks (ratio 4) for the world and the action tokens, for 504.0M backbone parameters. The mask acts on the scene ($\mathcal{S}$), the summary tokens ($\mathcal{M}$), and the actions ($\mathcal{A}$): scene tokens attend to $\mathcal{S}$, summary tokens to $\mathcal{S}$ and $\mathcal{M}$, and action tokens to $\mathcal{M}$ and $\mathcal{A}$. A blocked edge has the logit $-\infty$ in every layer and head, in training and at deployment. Because the world tokens attend neither to the actions nor to the flow time, their states are computed once per query and reused across the Euler steps.

A.4. Training Recipe

Table S1 lists the recipe on Dynamic-14. Every control step of a demonstration is a query window, which gives 1,324,374 training windows from the 11,200 demonstrations; 350 further episodes form the validation split. Each window contains 8 query positions with their own causal histories and no attention across positions. Each loss is normalized by its count of valid labels over the accumulated global batch. The DINOv3 features of future frames are cached once per episode. We chose the depth of the model among 12, 24, and 48 layers at width 1024 on development scenes at equal numbers of training samples (Fig. S1a).

Table S1. Training hyperparameters on Dynamic-14.

Item	Value
Backbone	24 layers, width 1024, 16 heads, FFN 4096, 504.0M parameters
Tokens	212 scene, 16 summary, 8 action
Encoder	frozen MAE ViT-B/16, 224×224 input
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-4}
Learning rate	2×10^{-4}, 5,174 warm-up updates from 0, cosine decay to 0
Batch, updates	global batch 512 windows, 20 epochs (51,740 updates)
Loss weights	$\lambda_P = \lambda_F = 1$, $\lambda_R = 0.1$, $\lambda_{\text{aux}} = 0.1$ per target
Precision	trainable weights FP32, frozen encoder BF16, gradient clipping at 0.1
Action noise	8 noise–time pairs per query position
Sampling	5 Euler steps for the action
Inference	FP32, world states cached within a query
Hardware	[N] RTX PRO 6000 GPUs, [X] hours per run

Table S2. **DROID pretraining data.** Yield of each stage and validity of the labels. Windows are causal observation windows with aligned labels, one per control step of the main camera. Flow labels count point–horizon slots.

	Train	Val.	All
Episodes associated	–	–	42,935
Episodes with windows	37,874	4,213	42,087
Windows	538,740	60,149	598,889
Valid actions (%)	—	—	96.88
Valid flow labels (%)	—	—	47.05
Future frames for DINO	—	—	3,425,218
Windows with moving objects (%)	—	—	—

A.5. Pretraining on DROID

Data. The real-robot base trains on DROID [36], recorded at 15 Hz, together with the 3D point tracks and optimized camera calibrations that PointWorld released for it [30]. We associate each released clip with its raw DROID episode by exact agreement of the joint sequence, end-effector poses, gripper states, commands, and camera identity, and exclude clips with a non-monotonic capture clock or an ambiguous association. The model sees the first exterior camera, its depth, and PointWorld's calibration of it. Of 42,935 associated episodes, 42,087 yield usable windows: 538,740 anchor windows from 37,874 training episodes and 60,149 from 4,213 validation episodes, with no shared episode (Table S2). All times are capture timestamps. The robot history uses the 8 most recent states at or before the query time, and the flow and DINOv3 targets keep PointWorld's record times rather than being interpolated to 50 ms horizons. Each anchor expands into 8 consecutive query positions with their own causal histories; the anchor carries the flow and DINOv3 labels, and the other positions carry action labels only. Robot states and commands are expressed in the base frame after the calibrated flange-to-TCP transform. We keep the 96.88% of action slots at which the collector issued a command, without a filter on magnitude or speed, so that the base is not biased toward slow motion.

What the base learns. DROID provides no object poses, so the base has no target-state tokens and no pose head. Its inputs are the current RGB-D frame and the robot history (204 scene tokens), with unavailable depth masked, and its objectives are the action loss (weight 1), the sparse 3D flow (weight 1), and the future DINOv3 features (weight 0.1). PointWorld's tracks record only visible positions, so the flow labels contain no negative visibility examples, and the visibility loss is off. Most objects in DROID move only when the robot moves them, so the base mainly learns metric geometry, the motion caused by the robot's own actions, and its coupling with the actions; the exogenous motion of our tasks and the target pose enter during fine-tuning.

Figure placeholder: Fig. S1 — model scale and pretraining

(a) Simulation model. y: development SR; x: 12, 24, and 48 layers at equal sample exposure; the selected configuration starred. **(b) DROID base.** x: training samples (log); y: validation flow endpoint error at the last horizon (left axis, mm) and action loss (right axis) for the 48-layer run; marker at the selected checkpoint.

Figure S1. Scale of the simulation model (a) and pretraining of the DROID base (b).

Flow labels. Each patch of the 14×14 grid follows one PointWorld track over the window. Among the tracks that start inside the patch, are initially visible, and have a valid positive depth, we prefer the one with the largest image motion whose 3D displacement agrees with its image motion, and otherwise take the track nearest to the patch center. This rule covers moving objects that a fixed center point would miss: on 18 segments with visible object motion, it placed a dynamic track inside the object in 13, against 5 for the center rule. The label is the track's displacement from its initial position, and patches without a valid track are invalid; 47.05% of the point–horizon slots carry a label. On image-static surfaces, the selected tracks move by [0.9] and [5.8] mm (median and 95th percentile), against [18] and [72] mm for flow lifted from 2D correspondences [53] with monocular metric depth [74], so we use the PointWorld tracks.

Recipe. The base uses 48 layers at width 1024 (1.02B trainable parameters) and trains for 20 epochs (21,060 updates, 2,106 warm-up updates) at a global batch size of 512 on 64 GPUs, with the optimizer and precision of Table S1 (Fig. S1b).

Fine-tuning. Each real-robot policy is initialized from the base and fine-tuned on the demonstrations of its task. The target-state tokens, the pose head, and the appearance heads that the base lacks start from fresh initialization, and the policy receives poses from the tracker of Appendix C.2. No real-robot policy is initialized from simulation weights, and the base sees no simulation data.

A.6. Deployment

Each query encodes the newest observation, runs the world stream once without its heads, and integrates the action stream on the cached world states. The chunk starts at the first control-period boundary at or after the query ends, and the next query starts at that boundary with a new observation. While a query runs, physics advances and the executor continues with the remainder of the previous chunk (Appendix B.3).

B. Dynamic-14 and Evaluation Protocol

B.1. Tasks

We built Dynamic-14 on MuJoCo and robosuite [71, 104]. It contains fourteen tasks in five families (Fig. S2). Table S3 lists their success conditions and Table S5 their scenario ranges. We will release the tasks, the accepted scenarios, the success checks, and the demonstrations.

Roles. Table S4 assigns the bodies of each task to the two roles of Sec. 3.1. The target is the body whose pose error enters a contact or containment tolerance ε, and the context consists of bodies that only need to be avoided with clearance m. A moving receptacle is therefore a target, since its pose decides containment. The wire of the passage tasks plays both roles: its pose at the entry is a target, because the wand must be aligned within 30 mm there, and its geometry along the loop is context, because any contact fails. The last column lists the error terms of Eq. 2 that each task exercises (Appendix E).

Terminal events. We check success at every 2 ms physics step, after the ordered requirements of Table S3, and an interrupted continuous condition resets its counter. Terminal events fall into five categories. *Invalid* events are corrupted simulator states. *Boundary* events are a workspace or downstream exit, a drop, or an expired goal window. *Safety* events are unsafe contact, an excessive rim impulse, wire contact, a release impulse, early robot contact, unsupported tray contact, or a striker trap. *Task violations* are a release outside the goal, a path deviation or backtracking, a premature release, a deflection without capture, tipping, a missing or extra impact, or an insufficient change of outgoing motion. The last category is the *timeout*. Each family evaluates its events in a fixed order per physics step and keeps the first terminal event.

B.2. Scenarios and Demonstrations

We sample continuous ranges uniformly and discrete choices with equal probability, and then apply physical-feasibility checks. Evaluation replays the stored accepted scenarios. Harmonic translation follows $\mathbf{p}(t) = \mathbf{c} + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0)$ with $\phi \sim \mathcal{U}[-\pi, \pi]$. Capture targets start at rest and move under gravity and friction on ramps of $[5, 15]^\circ$. We accept an occluded scene only if the target is visible for at least eight initial frames, fully occluded for 0.5–3.0 s, and visible again for at least three frames.

The training pool contains 11,200 demonstrations, 800 per task, generated by experts with privileged simulator state; the learned policies receive only the inputs of Sec. 3.2. We

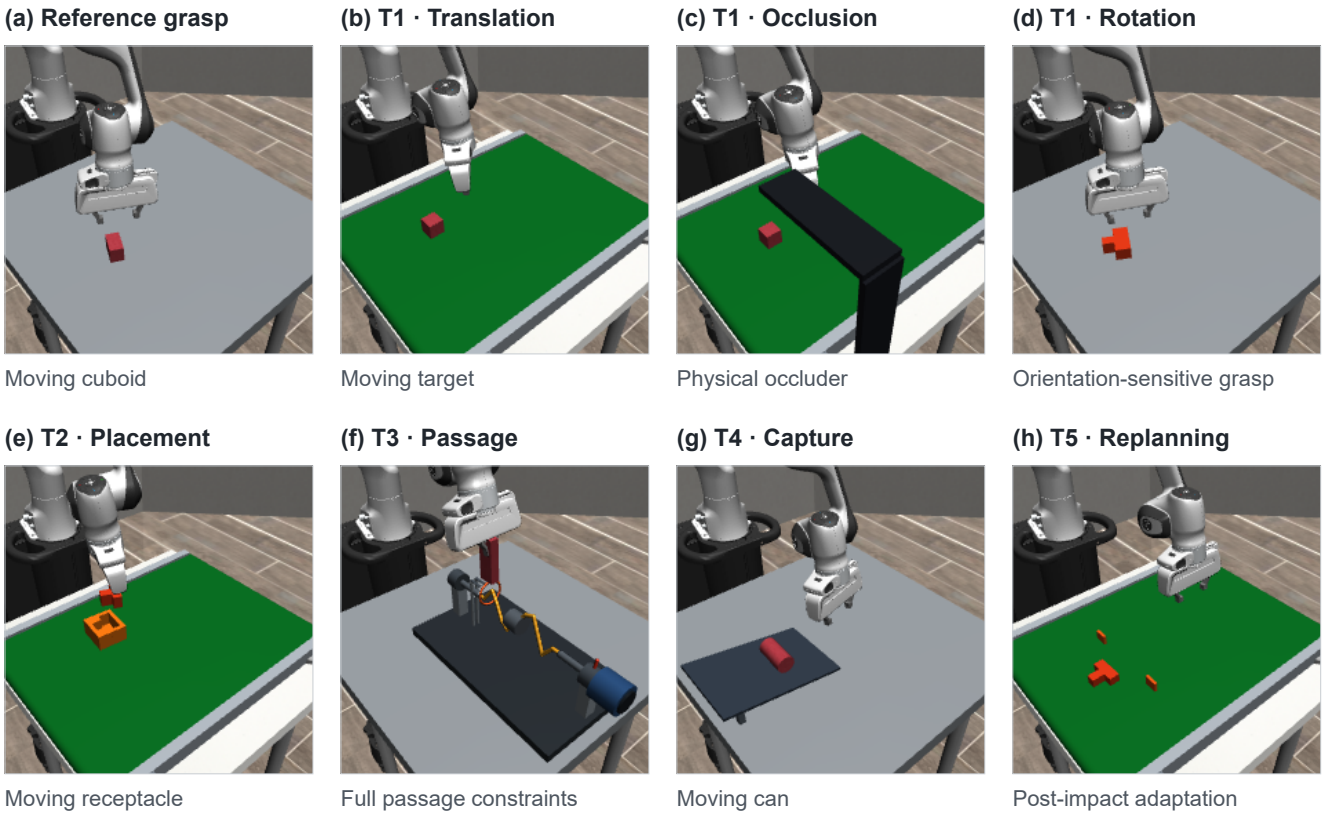


Figure S2. Dynamic-14 observations. Eight examples cover reference grasping, grasping a translating, occluded, or rotating target, placement into a moving receptacle, passage along a rotating wire, capture of a rolling object, and grasping after an impact. Each panel is the final observed, uncropped 224×224 agent-view RGB of one development scene. Table S3 lists all fourteen tasks.

record the demonstrations at 20 Hz with camera images, depth, robot states, issued commands, and simulator ground truth for target poses and scene geometry, from which we compute the future labels, their validity masks, and the command labels.

B.3. Execution Protocol

Executor. Every method runs under the same asynchronous executor on a 50 ms control clock, and queries start on its period boundaries. Let L_t be the total latency of query t, its inference time under the protocol plus any added delay. Its chunk is adopted after $k_t\Delta_a$ with $k_t = \max(1, \lceil L_t/\Delta_a \rceil)$, and the next query starts at that same boundary. While a query runs, the executor continues with the remainder of the previous chunk, one command per control period, and the new chunk replaces this remainder. The commands of query t that execute before replacement therefore execute at

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + (k_t + j)\Delta_a + d_{\mathrm{task}}, \qquad \text{(S1)}$$

for $j = 0, \ldots, \min(8, k_{t+1}) - 1$, where d_{task} is a command delay built into a task (50 ms for the post-impact grasp, zero otherwise). A method whose inference fits in one control period thus replans before every command. With $k_t = k_{t+1} = k \leq 4$, i.e., total latency up to 200 ms, the largest lead is $(2k-1)\Delta_a + d_{\mathrm{task}}$ and lies within the 400 ms of the forecast supervision; at longer latencies, the last commands of a chunk execute beyond it. Before the first chunk of an episode arrives, and whenever a chunk runs out before the next one, the controller holds the current arm target and gripper command while physics advances; no command is repeated.

Real-time protocol. Unless noted otherwise, physics advances during inference by a latency drawn per query from the method's measured latency distribution on one RTX 5090 (Appendix B.7). Table S13 reports three alternatives. With *paused* physics, the executor is unchanged and the simulation waits for inference; an added delay still advances physics. With *equal latency*, every method is charged the p95 latency of SE(3)-WAM regardless of its own speed. With *original executors*, each baseline keeps the execution scheme of its paper, e.g., a longer chunk before replanning.

Table S3. Success conditions of the fourteen Dynamic-14 tasks, with the thresholds of our task implementations. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

Task	Ordered success conditions	Limit
grasp; t1_belt, t1_xyz, t1_so3, t1_occ	Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC.	160 steps 8 s
t2_trans	Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both ≥ -0.002 m, top no higher than the opening. Release requires an opening command < -0.5 and no fingerpad contact. Relative linear/angular speeds ≤ 0.08 m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell.	200 10 s
t2_se3	Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance ≥ 0.002 m, floor gap in $[-0.002, 0.002]$ m, tilt ≤ 0.05 rad, top below opening. After release, relative speeds ≤ 0.02 m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach.	200 10 s
t3_phase	The episode starts with the wand already grasped. Phase alignment: distance ≤ 0.090 m and loop-direction error ≤ 0.20 rad. Enter within 0.030 m at progress ≤ 0.15; reach historical maximum progress ≥ 0.60, then ≥ 0.85 and endpoint distance ≤ 0.055 m.	420 21 s
t3_full	Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress ≥ 0.85 and move at least 0.045 m from both path and endpoint. Finally open (< -0.5) with no fingerpad contact. Releasing before exit fails.	640 32 s
t4_sphere	Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s.	120 6 s
t4_sphere_tabletop; t4_slider	Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed ≥ 0.08 m/s. Bilateral hold for 0.05 s, COM lift ≥ 0.05 m from the first attainment of that hold criterion, then contact and height for 0.50 s.	120 6 s
t4_can	Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical > 0.45 fails as unstable tipping. The configuration does not require a transverse closing axis.	120 6 s
t5_replan	Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed ≥ 0.035 m/s and yaw rate ≥ 0.35 rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift ≥ 0.05 m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s.	120 6 s

Added latency. An added delay of 0, 50, 100, 150, or 200 ms is inserted after inference on top of the protocol's latency, and physics advances during it.

Pose noise. With frozen weights, we evaluate three synthetic perturbations of the target pose: nominal temporally correlated noise, twice its amplitude, and nominal noise with contiguous missing observations. The nominal process combines a fixed per-episode bias and a stationary correlated component, each with per-axis standard deviations of 5 mm and 1°, with a correlation time of 200 ms. Dropouts last 200 ms, separated by geometrically distributed valid gaps with a mean of 3.8 s, giving a long-run missing fraction of 5%. The perturbation applies to the current and past estimates before the history features are computed.

B.4. Baselines

Common conditions. Table S6 summarizes the adaptations. Each baseline keeps its official architecture, inputs, action chunk, and initialization. ACT and Diffusion Policy train from scratch, $\pi_{0.5}$ is fine-tuned from `pi05_base`, and the other methods start from their released weights. All of them train on the same 11,200 demonstrations, whose commands we map to each action space through the same Cartesian controller. Dynamic-14 renders a 224×224 wrist camera in addition to the agent view, and methods whose recipes use a wrist view receive it; SE(3)-WAM and its CV-input variant use the agent view only. Each baseline follows its official recipe for the published number of updates, trained once with seed 0, and we change only the data loading and the action space. For DreamVLA, we ran both its original 40-epoch schedule and a 20-epoch cosine schedule

Table S4. **Roles and tolerances of the Dynamic-14 tasks.** The target is the body whose pose error enters a contact or containment tolerance ε. The context comprises bodies that must only be avoided, with clearance m. Speeds follow the sampling ranges of Table S5 (capture tasks: at the ramp exit). Terms: error terms that a task exercises (Eq. 2 and Appendix E). v: moving target. $r\theta$: orientation enters the contact. e_{dyn}: non-constant motion (harmonic, gravity, friction, impact). m: a context body must be avoided. Occ.: history gaps. Delay: command delay built into the task.

Task	Code	Target	Context	ε (mm)	$\|\mathbf{v}\|$ (m/s)	m (mm)	Terms
Harmonic grasp	grasp	block	table	[X]	0.03–0.20	–	v, $r\theta$, e_{dyn}
Conveyor grasp	t1_belt	part	belt	[X]	0.04–0.10	–	v
Translating grasp	t1_xyz	part	table	[X]	0.04–0.15	–	v, e_{dyn}
Rotating grasp	t1_so3	part ($\omega \leq 1.2$ rad/s)	table	[X]	0.04–0.15	–	v, $r\theta$, e_{dyn}
Occluded grasp	t1_occ	part	occluder canopy	[X]	0.04–0.10	[X]	v, m, occ.
Moving receptacle	t2_trans	receptacle	table	[X]	0.03–0.06	–	v
Rotating receptacle	t2_se3	T-shaped receptacle	table	[X]	0.03–0.06	–	v, $r\theta$
Wire passage	t3_phase	wire loop (entry)	rotating wire	30	[X]	swept	v, $r\theta$, m
Wire passage, full	t3_full	wire loop (entry)	rotating wire	30	[X]	swept	v, $r\theta$, m
Ball capture	t4_sphere	ball	tray, ramp	[X]	[0.4–0.7]	[X]	v, e_{dyn}, m
Ball capture, table	t4_sphere_tabletop	ball	table, ramp	[X]	[0.4–0.7]	–	v, e_{dyn}
Slider capture	t4_slider	slider	table, ramp	[X]	[0.4–0.8]	–	v, e_{dyn}
Can capture	t4_can	can	table, ramp	[X]	[0.4–0.7]	–	v, $r\theta$, e_{dyn}
Post-impact grasp	t5_replan	part	striker	[X]	0.12–0.14	[X]	v, $r\theta$, e_{dyn}, m, delay

and report the one that was better on development scenes.

Inputs. PUMA [19] receives its historical optical flow, computed on the rendered frames. DreamVLA receives seven causal RGB frames, the end-effector position and orientation, the finger positions, and a fixed task text, and it samples with ten DDIM steps and classifier-free guidance of 1.5. OpenWAM-α receives the current RGB views, joint values, gripper opening, and task text, and it generates video and actions jointly with video decoding disabled. DynamicWAM receives its flow history and kinematic descriptors.

SE(3)-augmented baselines. $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Appendix A.1 from the same pose source, with the same validity flags and masking. A two-layer MLP maps the tokens to the model's width. For $\pi_{0.5}$, we append them to the prefix after the image tokens, where the action expert attends to them; for DynamicWAM, they join its kinematic descriptors. The output layer of the MLP is zero-initialized, and training follows the base model's recipe.

CV-input variant. *SE(3)-WAM, CV input* shares the scene tokens and the action stream of SE(3)-WAM but has no forecast losses. Its action stream instead receives constant-velocity extrapolations of the target pose at the chunk's estimated execution times as input tokens, computed from the history and a delay estimate available at query time. It is trained on the same data with the same recipe and keeps the current-target RGB reconstruction.

B.5. Splits and Statistics

We select configurations on 1,400 development scenes (100 per task). All Dynamic-14 results use a separate held-out set of 1,400 scenes generated with the same ranges and new source seeds that appear in no training, validation, or development episode; the first 100 feasible scenes in increasing seed order form each task's set. The full model, the CV-input variant, and the four ablation controls use training seeds 0, 1, and 2, and each baseline uses seed 0. On development scenes, three training seeds of the same configuration differed by up to [7.4] points in SR, mostly on the wire passage, whose success ranged from [0] to [71]% across seeds. We report each training seed and the cross-seed mean and standard deviation. For paired 95% intervals, we average the seeds per scene and bootstrap held-out scenes within each task, jointly across methods. These intervals quantify scene uncertainty for the trained weights, and we treat differences whose interval lies within ± 2 points as ties.

B.6. Metrics

The task-balanced success rate is

$$\mathrm{SR}(m) = \frac{1}{14} \sum_{k=1}^{14} \frac{1}{n_k} \sum_{i=1}^{n_k} y_{mki}, \tag{S2}$$

for binary outcomes y_{mki} of method m on scene i of task k. The *position error at first contact* e_p is the distance of the end effector from the contact frame at the first contact. The *contact-time error* e_t is the distance from the first-contact time to the feasible contact window of the scene, and it is zero inside the window. The window is the set of control-period boundaries t at which the privileged expert could make the contact of the task's first success stage. A time t

Table S5. Scenario proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

Task	Initial-state proposal	Motion or event parameters
grasp	Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$.	$v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$.
t1_belt	Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist.	Belt $v_y \in [.04, .10]$.
t1_occ	Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$.	Same belt range; clearance parameter .20; physical occlusion.
t1_xyz	Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$.	$v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$.
t1_so3	Same as XYZ.	Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities.
t2_trans	Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation.	Goal $v_y \in [.03, .06]$; no rotation.
t2_se3	Same goal x, y range, $z = .804$; yaw $[-.3, .3]$.	Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities.
t3_phase	Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$.	Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values).
t3_full	Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range.	Same wire rotation; cradle and release requirement enabled.
t4_sphere	Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$.	Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture.
t4_sphere_tabletop	Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$.	Same slope and release; target/ramp friction $[.7, .9]$, table friction .35.
t4_slider	Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp.	Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$.
t4_can	Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent.	Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$.
t5_replan	$x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist.	Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms.

belongs to it when the expert's contact frame on the target at t, taken from a rollout of the same scene with the robot held at its initial pose, (i) satisfies the geometric conditions of that stage (Table S3), (ii) has an inverse-kinematics solution within the expert's joint limits, and (iii) can be reached from the initial end-effector pose by t under the expert's per-command limits of 0.05 m per translation axis and 0.5 rad per rotation-vector component. The earliest such time is the larger of $\lceil \max_i |\Delta p_i| / 0.05\,\mathrm{m} \rceil$ and $\lceil \max_i |\Delta \phi_i| / 0.5\,\mathrm{rad} \rceil$ control periods, a lower bound that ignores the controller's tracking lag. In this reference rollout, only robot–target collisions are disabled. The window depends only on the scene and is shared by all methods. The two placement tasks start with the object held, so their contact is the first contact of the held object with the receptacle, and their contact frame is the end-effector pose implied by the expert's placement target. The phase-only passage (`t3_phase`) starts with the wand held and succeeds without any contact, so the averages of e_t and e_p cover the other 13 tasks, while SR covers all 14. We report both errors on episodes that make contact; failures without contact stay in the SR denominator. *Time-to-action* is the wall-clock time from the arrival of an observation to the availability of the first command, and the *action age* is the execution time minus the capture time.

B.7. Timing

We time all methods on one RTX 5090 (32 GB), the deployment GPU of the robot. For each method, each of three rounds uses thirty warm-up calls and 300 measured calls over three observation sequences of eight frames, with CUDA

Table S6. **Baseline adaptations.** All methods train on the same 11,200 demonstrations with the inputs of their papers and run under the shared executor of Appendix B.3. Updates × batch follows the official recipe. Exec.: the execution scheme of the original paper (commands executed per chunk before replanning), used only for the original-executor control of Table S13. Params: trainable parameters, frozen encoders excluded. Seeds: training seeds.

Method	Initialization	Inputs	Updates × batch	Exec.	Params	Seeds
ACT [98]	scratch	agent + wrist RGB, robot state	50,000 × 32	temporal agg.	[X]	1
Diffusion Policy [16]	scratch	agent + wrist RGB (published horizon), robot state	50,000 × 32	8	[X]	1
$\pi_{0.5}$ [56]	`pi05_base`	current agent + wrist RGB, robot state, task text	30,000 × 256	10	[X]	1
$\pi_{0.5}$ + SE(3)	`pi05_base`	as $\pi_{0.5}$ + SE(3) motion tokens	30,000 × 256	10	[X]	1
DynamicVLA [81]	released DOM ckpt.	frames $[-2, 0]$, robot state, task text	50,000 × 44	streaming	[X]	1
PUMA [19]	official checkpoint	RGB, optical-flow history, robot state, task text	[X]×[X]	[X]	[X]	1
DreamVLA [94]	released components	seven causal agent + wrist RGB frames, robot state, task text	48,220 × 512	1	[X]	1
OpenWAM-α [75]	official foundation	current agent + wrist RGB, joints, gripper, task text	34,490 × 192	32	[X]	1
DynamicWAM [49]	official initialization	RGB, flow history, kinematic descriptors	40,000 × 128	16	[X]	1
DynamicWAM + SE(3)	official initialization	as DynamicWAM + SE(3) motion tokens	40,000 × 128	16	[X]	1
SE(3)-WAM, CV input	as SE(3)-WAM	as SE(3)-WAM, plus constant-velocity forecast tokens, no forecast losses	as SE(3)-WAM	–	[X]	3
SE(3)-WAM	frozen MAE encoder	current agent RGB-D, SE(3) and robot histories	51,740 × 512	1	523.8M	3

Table S7. **Latency breakdown** of one query of SE(3)-WAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot.

Stage	p50	p95
Pose tracking (per frame, robot)	—	—
Image encoding	—	—
World stream (once per query)	—	—
Action stream (5 Euler steps)	—	—
Time-to-action (total)	—	—
Forecast heads (training only, for reference)	—	—
Action age at execution (robot)	—	—
Queries over 50 ms (robot, %)	—	

synchronization at call boundaries and a rotated method order. Each method uses the precision of its release; SE(3)-WAM uses FP32. The time-to-action includes encoding the image and generating the first command, and on the robot also tracking the newest frame. For SE(3)-WAM, we time the forecast heads separately, although deployment skips them. Table 1 gives the p95 values and Table S6 the parameter counts. On the robot, traces record capture, availability, preprocessing, the start and end of inference, queue entry, command issue, and execution. We report the p50, p95, and maximum action age. A query misses its deadline when its time-to-action exceeds the 50 ms control period, so that the next period executes a remaining command or holds; we report the fraction of all queries of the test trials that miss it, together with the peak memory (Table S7).

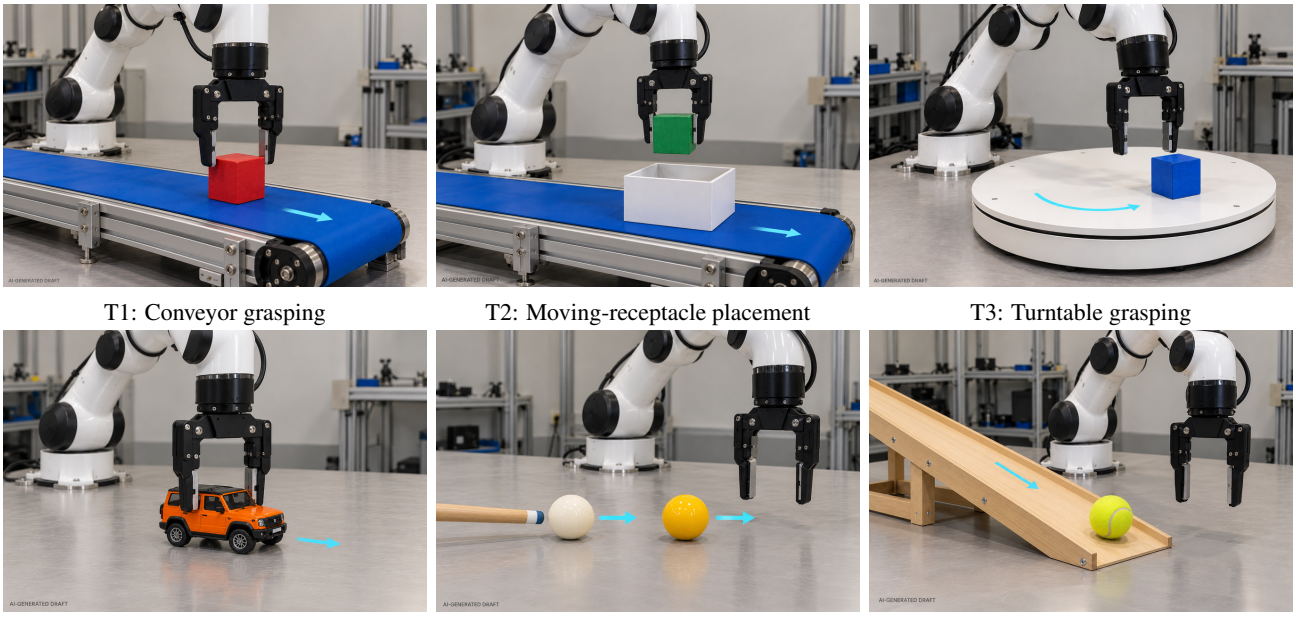
C. Real-Robot Setup

C.1. Platform and Tasks

The robot is a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera, commanded through [a timestamped Cartesian interface] at [control frequency]. The camera records RGB-D at [resolution] and [frame rate], and we calibrate its extrinsics by [procedure] before each session. SE(3)-WAM uses this fixed third-person view; baselines whose recipes require a wrist view receive [per-method camera configuration]. All policies run on one RTX 5090. Table S8 defines the six tasks. T1 and T2 share a belt conveyor, T3 uses a turntable, T4 a remotely driven toy car, T5 a cue-driven collision between two balls, and T6 a ramp. Touching or pushing the target without completing the task, a single-finger hold, and an action after the target has left the workspace count as failures; [equipment-invalid trials and safety interventions are repeated or counted as failures by the following rule].

C.2. Pose Tracking

Tracking is marker-free. Each object has one fixed reference model with a shared origin, axes, and scale across all demonstrations and trials. Grounding DINO [44] localizes the object named by the task in the first frame, SAM 2.1 [59] propagates its mask causally, and [the tracker, e.g., FoundationPose [76]] registers the masked RGB-D observation to the reference. [Image resolution, reference acquisition, and loss and recovery thresholds.] The tracker runs in its own thread. Each estimate carries its capture time, availability time, and validity, is issued once, and is never revised by later frames. Training labels for the demonstrations use the same object frame with multiframe registration. For the ball



T1: Conveyor grasping | T2: Moving-receptacle placement | T3: Turntable grasping

T4: Moving-car grasping | T5: Post-collision interception | T6: Ramp interception

Figure S3. **Real-robot tasks.** AI-generated concept illustrations, to be replaced by photographs of the setup; arrows indicate the intended target motion. T6 shows the rolling variant; the task also includes sliding objects.

Table S8. Real-robot task definitions. $N = 100$: training demonstrations per task; 20 additional demonstrations per task are reserved for validation.

Task	Device, motion, and objects	Operating range; N	Success and failure criteria
T1	Conveyor transports a graspable cube.	[belt speed]; 100	[stable grasp and lift; failures]
T2	Conveyor transports an open box; the arm places a cube into it.	[belt speed]; 100	[placement and retention; failures]
T3	Turntable carries a cube around its axis.	[angular speed, radius]; 100	[stable grasp and lift; failures]
T4	Remotely driven toy car moves on the tabletop.	[speed, path]; 100	[car grasp and lift; failures]
T5	A cue-driven ball strikes a second ball; the arm intercepts the second ball after the collision.	[impact and outgoing speed]; 100	[post-impact capture; failures]
T6	A ball rolls or a block slides down a ramp toward the interception region.	[slope, release, speed]; 100	[capture and retention; failures]

Table S9. **Real-robot pose tracking.** Median/p95 translation (mm) and rotation (°) error of the tracker against a marker-free reference, the uncertainty of that reference, and the fraction of lost observations. Ball rotation is not used.

Target	mm	°	Ref. uncertainty	Lost
Real rigid object	—	—	—	—
Real ball	—	–	—	—

tasks, the target-state tokens keep the translation, velocity, and acceleration of the ball center and mask its orientation and angular motion, and the pose head is supervised on translation only. Table S9 reports the tracking accuracy.

C.3. Trials

We compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [16], with 20 trials per task and method. All methods use the same pre-generated trial keys, which fix the object instance, the initial pose, and the motion parameters, sampled by [sampling protocol]. Trials run in [randomized method order], so that drift of the apparatus does not align with the methods. An operator resets the scene. A trial starts at the first usable observation, the robot may move at any time, and a trial ends at success, failure, or a timeout of [X] s. Demonstrations, development trials, and test trials come from separate sessions. SE(3)-WAM is fine-tuned per task from the DROID base (Appendix A.5), and the baselines are trained or fine-tuned on the same demonstrations with [per-method initialization and training settings]. Table S10

reports successes per task, 95% binomial intervals for the pooled rates, and paired differences from a bootstrap over trial keys.

Table S10. **Real-robot successes** per task (successes/trials, 20 trials per task; tasks in Table S8). All: pooled success rate (%) with its 95% binomial interval. Δ: paired difference from SE(3)-WAM (points) with a bootstrap interval over trial keys.

Method	T1	T2	T3	T4	T5	T6	All	Δ
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	–

D. Additional Results

D.1. Per-Task Results and Outcomes

Table S11 breaks the held-out Dynamic-14 results of Table 1 down by task for every method, with the paired interval of SE(3)-WAM against the strongest other method on each task and the contact-time error of each method. Table S12 reports how episodes end, so that a method that trades one failure for another, e.g., fewer collisions for more timeouts, becomes visible. Every failure stays in the SR denominator.

The SE(3) placement into the rotating receptacle (t2_se3) fails for every method (Table S11): [the policies reach the receptacle but miss its angular tolerance before it turns away]. No variant of SE(3)-WAM solves the full wire passage (t3_full), which adds grasping and releasing the wand to the passage, whereas [OpenWAM-α] succeeds in [XX]% of these episodes. Most remaining failures of SE(3)-WAM are [e.g., timeouts after a missed first grasp].

D.2. Protocols and Latency

Table S13 evaluates every method under three further protocols. Paused physics removes the cost of computation, equal latency charges every method the p95 latency of SE(3)-WAM and so separates a method's speed from the way it uses time, and the original executors test whether the shared executor disadvantages any method. With paused physics, the difference between SE(3)-WAM and [DreamVLA] is [±X.X] points, and under equal latency it is [±X.X] points. The original executors change the SR of the baselines by at most [X.X] points. Table S14 gives the success at every added latency; its rows for SE(3)-WAM and the strongest baseline are the data of Fig. 1 (right).

D.3. DOMINO

On DOMINO [19], we compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of its clean level-1 setting. All methods use the benchmark's demonstrations, scenes, and native evaluator, one training seed, and 50 episodes per task and method. We report SR and the manipulation score (MS), which scales route completion by penalties for leaving the workspace or view and for collisions (Table S15). SE(3)-WAM uses domain-specific normalization and reads the simulator pose of the target named by the instruction; slots without a valid pose are masked as in Appendix A.1. The RGB-based baselines do not receive this pose. [Summarize the DOMINO results.]

Figure placeholder: Fig. S4 — pose noise
x: clean poses, nominal correlated noise, twice its amplitude, and nominal noise with missing observations; y: SR of SE(3)-WAM per task family, frozen weights, real-time protocol; paired intervals over matched scenes.

Figure S4. **Robustness to pose noise.** SR of SE(3)-WAM under synthetic perturbations of the target pose, relative to clean poses.

D.4. Pose Noise

Fig. S4 evaluates the final model with frozen weights under the three synthetic pose perturbations of Appendix B.3, on the same held-out scenes at zero added latency and with all three training seeds. Relative to clean poses, SR changes by [X.X], [X.X], and [X.X] points under nominal correlated noise, twice its amplitude, and missing observations.

D.5. Design Screening

We screened the encoder, the history representation, the forecast targets, and the coupling of world and action streams with smaller models: 4 layers of width 768, trained for 4 epochs (40,104 updates at a global batch size of 16) and evaluated with the synchronous executor that preceded our real-time protocol. Each group keeps its own encoder and seed count, stated below, and conditions are compared within a group. These studies selected the components of the final model; the four ablations of Sec. 4.4 then test its central components at full scale.

Encoder and backbone. We compared a CNN trained with the policy, frozen DINOv3 ViT-S/16 and ViT-L/16 [63], and MAE ViT-B/16 [26] (three seeds for the MAE reference; Table S16), and selected MAE. We also compared a backbone with shared attention and stream-specific feed-forward networks against one with fully separate world and action parameters, at about 47.3M backbone parameters and

Table S11. **Per-task held-out success** (%) on Dynamic-14 for every method of Table 1, mean of three training seeds for our variants and one for baselines, real-time protocol. DP: Diffusion Policy. DVLA: DynamicVLA. DrVLA: DreamVLA. OWAM: OpenWAM-α. DWAM: DynamicWAM. +: with our SE(3) history tokens. CV: SE(3)-WAM with CV input. σ: SE(3)-WAM with nominal synthetic correlated pose noise (Appendix D.4). Δ: SE(3)-WAM minus the strongest other method on that task, with its 95% paired interval. The last row gives the contact-time error e_t (Appendix B.6). Task codes are listed in Table S4.

Task	ACT	DP	$\pi_{0.5}$	$\pi_{0.5}$+	DVLA	PUMA	DrVLA	OWAM	DWAM	DWAM+	CV	σ	**Ours**	Δ [95% CI]
Harmonic grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Conveyor grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Translating grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Rotating grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Occluded grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Moving receptacle	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Rotating receptacle	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Wire passage	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Wire passage, full	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ball capture	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ball capture, table	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Slider capture	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Can capture	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Post-impact grasp	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Task-balanced	—	—	—	—	—	—	—	—	—	—	—	—	—	—
e_t (ms)	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table S12. **Outcome categories** on held-out Dynamic-14 scenes (% of episodes, seed-averaged, real-time protocol). Categories follow Appendix B.1, and rows sum to 100. The noisy-pose row reuses the nominal synthetic correlated-noise episodes from the robustness study.

Method	Success	Safety	Task	Bound.	Timeout	Inval.
ACT	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—	—
DynamicVLA	—	—	—	—	—	—
PUMA	—	—	—	—	—	—
DreamVLA	—	—	—	—	—	—
OpenWAM-α	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—	—
SE(3)-WAM, CV input	—	—	—	—	—	—
SE(3)-WAM, noisy poses	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—

Table S13. **Protocol controls** on held-out Dynamic-14 scenes (SR, %). Paused: shared executor, with physics stopped during inference. Equal latency: shared executor, with every method charged the p95 latency of SE(3)-WAM. Original executor: real time with each method's own execution scheme. Real time: the protocol of Table 1. Late: queries in real time whose time-to-action exceeds the 50 ms control period (%).

Method	Paused	Equal latency	Original executor	Real time	Late (%)
ACT	—	—	—	—	—
Diffusion Policy	—	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—
DynamicVLA	—	—	—	—	—
PUMA	—	—	—	—	—
DreamVLA	—	—	—	—	—
OpenWAM-α	—	—	—	—	—
DynamicWAM	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—
SE(3)-WAM, CV input	—	—	–	—	—
SE(3)-WAM	—	—	–	—	—

with three seeds each (Table S22), and selected the shared attention.

History. With a frozen DINO-L encoder and one seed per condition, we compared the current state alone, the target-motion history, RGB windows, Farnebäck flow [20], flow with motion statistics [49], codec motion vectors, video-VAE latents, and compressed windows (Table S17). Table S18 replaces the past target poses of the trained history model by its current pose at test time.

Forecast targets. With a frozen DINO-L encoder and one seed per condition, we compared fifteen future targets (Table S19), from target-only and image-space forecasts to role-

Table S14. **Success versus added latency** on held-out Dynamic-14 scenes (SR in %, real-time protocol). The 0 and 150 ms columns equal the All and +150 ms columns of Table 1. Slope: least-squares change of SR per 100 ms of added latency (points).

	Added latency (ms)					
Method	0	50	100	150	200	Slope
ACT	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—	—
DynamicVLA	—	—	—	—	—	—
PUMA	—	—	—	—	—	—
DreamVLA	—	—	—	—	—	—
OpenWAM-α	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—	—
SE(3)-WAM, CV input	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—

aware combinations (Table S20). A second group with the MAE encoder and one seed removed each future appearance target from the selected configuration. Table S21 compares the learned SE(3) forecast of the final model offline with constant-velocity and constant-acceleration extrapolation from the same pose history.

Coupling. With the MAE encoder and three seeds per condition, we compared six couplings of the world and action streams. R0 disables every forecast loss. R1 adds one object query and one query per patch to the world stream, which attend to $\mathcal{S}$, $\mathcal{M}$, and each other and carry the forecast readouts, without any edge to the actions. R2 lets the action tokens attend to these queries at every layer, R3 lets the queries attend to the actions, and R4 opens both edges; in R3 and R4, the world states are recomputed at every Euler step. R5 decodes the forecast once per query and encodes the target trajectory and the flow with visibility, with rotations in a continuous 6D form, through a geometry encoder (0.37M parameters) into tokens that the actions attend to. Feeding forecasts to the actions gave no consistent gain: R1 exceeded R0 by [3.14] points on average, with seed-wise differences of [−1.29], [+9.64], and [+1.07], and the couplings that read or decode forecasts changed sign across seeds (Table S22). We therefore use the forecasts only as training supervision, which also keeps deployment free of their cost.

D.6. Forecasts at Execution Time

Fig. S5 aligns the query, the forecast, the ground truth, the issued actions, and the first contact in physical time. The examples show where forecast errors arise, e.g., at an impact or when a rolling object leaves its ramp.

D.7. Supplementary Video

The supplementary video shows real-robot trials of SE(3)-WAM and the strongest baseline on all six tasks with the tracked poses overlaid, Dynamic-14 episodes of every task family with the forecasts of the training-time heads, and failure cases.

Table S15. **DOMINO evaluation** on 24 single-arm clean-L1 tasks (one training seed per method, 50 episodes per task and method). SR (%) is grouped by interaction type and averaged over tasks; MS is the manipulation score. Protocol: Appendix D.3.

Method	Intercept	Track	All	MS
$\pi_{0.5}$	—	—	—	—
DynamicWAM	—	—	—	—
PUMA	—	—	—	—
SE(3)-WAM	—	—	—	—

Figure placeholder: Fig. S5 — forecasts aligned with what happened

Rows 1–4: the median-error held-out episode of each motion type (conveyor, rotating target, rolling capture, post-impact); **row 5:** a failure. **Columns:** the query frame at τ_t; the frame at the first command's execution time τ^{a}; and a 3D inset in the base frame. **Overlays:** blue past SE(3) poses; orange predicted target triads at $\tau_t + \delta_h$ and sparse flow arrows, read from the training-time heads; gray ground-truth poses at the same physical times; purple issued gripper path; a marker at first contact. A strip under each row plots translation and rotation error vs. horizon for learned SE(3), constant velocity, and constant acceleration. Annotate row 5 in red.

Figure S5. Forecasts of the training-time heads on held-out episodes, aligned in physical time with the ground truth and the issued actions. Rows 1–4 show the median-error episode of each motion type, and row 5 shows a failure.

Table S16. **Observation encoders** at the screening scale (Appendix D.5). Params: encoder parameters; p95: time-to-action.

Encoder	Params	Trained	SR	SR_{150}	p95
CNN	—	yes	—	—	—
DINOv3 ViT-S/16	22M	frozen	—	—	—
DINOv3 ViT-L/16	300M	frozen	—	—	—
MAE ViT-B/16*	86M	frozen	—	—	—

Table S17. **History representations** at the screening scale (frozen DINO-L encoder, one training seed, 1,400 scenes). The first eight conditions receive the current target pose; the last adds the target-motion history.

History input	Success / 1,400	SR (%)
Current image	748	53.43
RGB window	662	47.29
Optical flow	720	51.43
Flow + motion statistics	723	51.64
Codec motion vectors	733	52.36
Video-VAE latent	710	50.71
Compressed window (age)	634	45.29
Compressed window (uniform)	670	47.86
Current image + target-motion history	824	58.86

Table S18. **Stationary-history intervention** on the history model of Table S17 (1,400 scenes). At test time, past target poses are replaced by the current pose and the kinematic features are recomputed.

Target history	Success / 1,400	SR (%)
Unchanged	824	58.86
Stationary replacement	9	0.64

Table S19. Future-target definitions of the forecast screening. Shape excludes batch and query axes. Displacements are cumulative from the current state, except for codec vectors, which keep the correspondence of the decoder. All targets come from the recorded demonstrations. Future depth, DINO, and dynamic RGB are also the training-only auxiliary targets of the selected model.

Target	Shape	Support, units, and validity
Target pose	8×6	Cumulative translation (m) and left-relative rotation vector (rad). Requires a valid pose and query.
Scene flow, 14×14 grid	$196 \times 8 \times 3$	Fixed source grid, cumulative displacement in base-frame meters. Requires valid source depth and a finite future displacement.
Target tracks (2D)	$32 \times 8 \times 3$	Target-mask source points with two image-displacement channels and visibility; padded points are invalid. This is the P09 target, not a 3D rigid-pose fit.
Object DINO	8×1024	Mask-weighted pooling of future patch features with channel layer normalization. Requires a positive mask area.
Scene tracks (2D)	$196 \times 8 \times 3$	Grid source points with two image displacements divided by 224 and a visibility logit.
Dense 2D displacement	$8 \times 2 \times 224 \times 224$	Fixed source pixels, displacement divided by 224. Requires a valid source and a finite projection.
Codec vectors	$8 \times 2 \times 64 \times 64$	Each future/current pair is encoded as P/I frames, and vectors are divided by 64. Only valid vector regions count.
Video-VAE latent	$48 \times 2 \times 14 \times 14$	Frozen Wan2.2 VAE posterior mean with the VAE's own normalization. The two groups end at 200 and 400 ms.
Future depth	$8 \times 1 \times 224 \times 224$	Metric depth on finite positive pixels.
Future DINO	$8 \times 196 \times 1024$	Frozen DINOv3 ViT-L/16 teacher [63], 14×14 patches, channel layer normalization.
Dynamic RGB	$8 \times 3 \times 224 \times 224$	RGB divided by 255 on surfaces moving more than 1 mm, including the robot.

Table S20. **Forecast targets** at the screening scale. The first group uses a frozen DINO-L encoder and seed 0, with its 140 development scenes and the 1,260 remaining scenes reported separately. The second group uses the MAE encoder and seed 1 on 1,400 scenes; it keeps scene flow, SE(3), and the current-target RGB reconstruction and removes future appearance targets. Compare conditions within each group.

ID	Future supervision	Dev. success / 140	Success / 1,260	SR_{1260} (%)
P00	None	88	791	62.78
P01	3D scene flow	95	782	62.06
P02	SE(3)	92	789	62.62
P03	Scene flow + SE(3)	93	786	62.38
P04	Video-VAE latent + SE(3)	92	783	62.14
P05	Dense 2D displacement + SE(3)	87	746	59.21
P06	Codec motion + SE(3)	86	795	63.10
P07	Sparse scene 2D tracks + SE(3)	90	807	64.05
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target 2D tracks	88	808	64.13
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65
AUX0 MAE-B: future appearance auxiliaries		Success / 1,400		SR_{1400} (%)
None		894		63.86
Depth only		867		61.93
DINO only		946		67.57
Dynamic RGB only		926		66.14
All three (existing reference)		995		71.07

Table S21. **Offline prediction errors.** Target-position error at 400 ms (mm) on the same held-out observations. CV and CA are analytic extrapolations from causal pose history; the learned head reads frozen full-model states. These diagnostics are distinct from the trained CV-input policy in Table 1.

	Const. vel.	Harmonic	Roll/slide	Impact
CV extrapolation	—	—	—	—
CA extrapolation	—	—	—	—
Learned pose head	—	—	—	—

Table S22. **Couplings R0–R5 and backbones** at the screening scale (MAE encoder, three training seeds each; Appendix D.5).

Coupling	SR mean	SR std.	p95 (ms)
R0 (future losses off)	—	—	—
R1 (queries, no edge)	—	—	—
R2 (world→action)	—	—	—
R3 (action→world)	—	—	—
R4 (bidirectional)	—	—	—
R5 (decoded geometry)	—	—	—

E. Error Decomposition

This appendix derives the decomposition of Eq. 2 that motivates the design of SE(3)-WAM (Sec. 3.1). It describes how the errors of an estimate scale with the lead and the motion; it is not a model of success.

E.1. Translation

Let the true target position at the execution time be $\mathbf{p}(\tau^{\mathrm{a}}) = \mathbf{p}(\tau_t) + \mathbf{v}\Delta + \boldsymbol{\rho}$, where $\mathbf{v}$ is the velocity at τ_t and $\boldsymbol{\rho}$ the remaining motion, and let the policy's estimate be $\hat{\mathbf{p}}(\tau^{\mathrm{a}}) = \hat{\mathbf{p}}(\tau_t) + \hat{\mathbf{v}}\hat{\Delta} + \hat{\boldsymbol{\rho}}$. Subtracting and adding $\mathbf{v}\hat{\Delta}$ gives the identity

$$\begin{aligned}\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}}) = \big[\hat{\mathbf{p}}(\tau_t) - \mathbf{p}(\tau_t)\big] + (\hat{\mathbf{v}} - \mathbf{v})\hat{\Delta} \\ + \mathbf{v}(\hat{\Delta} - \Delta) + (\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}),\end{aligned} \tag{S3}$$

and the triangle inequality yields Eq. 2 with $e_v = \|\hat{\mathbf{v}} - \mathbf{v}\|$ and $e_{\mathrm{dyn}} = \|\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}\|$. The residual $\boldsymbol{\rho}$ is exactly the part of the motion that constant-velocity extrapolation misses: under a constant acceleration $\mathbf{a}$, $\boldsymbol{\rho} = \frac{1}{2}\mathbf{a}\Delta^2$, so an acceleration estimate from the history removes part of e_{dyn}, whereas impacts, rolling contact, and friction changes leave a remainder that only a model of the dynamics can reduce. For a learned predictor, the split of its estimate into $\hat{\mathbf{v}}\hat{\Delta}$ and $\hat{\boldsymbol{\rho}}$ is not unique, so the decomposition states which information each term requires rather than how a particular network uses it.

E.2. Rotation and the Contact Point

With $R(\tau^{\mathrm{a}}) = \mathrm{Exp}(\boldsymbol{\omega}\Delta + \boldsymbol{\eta})R(\tau_t)$ and $\hat{R}(\tau^{\mathrm{a}}) = \mathrm{Exp}(\hat{\boldsymbol{\omega}}\hat{\Delta} + \hat{\boldsymbol{\eta}})\hat{R}(\tau_t)$, rotations about a common fixed axis compose additively, and

$$\theta = \big\| \operatorname{Log}(\hat{R}(\tau^{\mathrm{a}})R(\tau^{\mathrm{a}})^{\top})^{\vee} \big\| \le \theta_{\mathrm{loc}} + \hat{\Delta}\, e_\omega + \|\boldsymbol{\omega}\|\, e_\tau + e_\eta, \tag{S4}$$

with the current orientation error θ_{loc} and e_ω and e_η defined as for translation; for general rotations, the Baker–Campbell–Hausdorff formula adds terms of second order in the angles. A contact point at $\mathbf{c}$ in the object frame, with $\|\mathbf{c}\| = r$, has the world position $\mathbf{p} + R\mathbf{c}$, so its error is $(\hat{\mathbf{p}} - \mathbf{p}) + (\hat{R}R^{\top} - I)R\mathbf{c}$. Since the operator norm of $\hat{R}R^{\top} - I$ for a rotation by θ is $2\sin(\theta/2)$,

$$e_{\mathrm{c}} \le \|\hat{\mathbf{p}} - \mathbf{p}\| + 2r\sin(\theta/2) \le \|\hat{\mathbf{p}} - \mathbf{p}\| + r\theta. \tag{S5}$$

Orientation errors therefore enter the contact scaled by the lever arm, and a grasp or placement that depends on orientation needs the full SE(3) state of the target.

E.3. Context and Clearance

Suppose that the executed path keeps every point of the robot and the carried object at least m away from the *predicted* positions of the context points it passes. A collision at a context point then requires the true position of that point to differ from its prediction by at least m. Under this assumption, context errors below the clearance do not cause collisions, which is why a coarse estimate of the surroundings can suffice when $m \gg \varepsilon$. The assumption concerns the executed path: SE(3)-WAM does not enforce a margin explicitly, so the effect of coarse context supervision is tested empirically by the ablations (Sec. 4.4). A flow sampled on a grid can also miss structures thinner than its spacing.

E.4. Scope

The decomposition addresses the approach to a rigid target whose motion is exogenous until contact. It does not cover deformable or articulated targets, whose motion is not determined by six numbers, or the motion after contact, which depends on the action itself and would call for action-conditioned prediction. Scenes in which the role of a body changes over time, e.g., sequential multi-object tasks, are also outside its scope.

F. Additional Related Work

This appendix extends Sec. 2 with work that the main text covers only briefly.

Dynamic manipulation and latency. Classical systems compensate latency with predictive tracking [4] and visual servoing [32], and catch or intercept moving objects by planning on estimated trajectories [2, 6, 27], later combined with grasp and reachability planning [1, 52]. Learned visuomotor policies [16, 98] and vision–language–action models [9, 11, 37, 55, 56, 106] generalize across objects and instructions, and most of them are trained and evaluated on scenes that stay still while the model computes. Among learned methods for moving scenes, DynamicManip synthesizes dynamic demonstrations from a single static one [41], and DynaWM refines the action chunk of a base VLA for moving objects [14].

World models and world–action models. World models have been used for planning with visual foresight [21], for reinforcement learning in a learned latent space [24, 25], and for planning on pretrained visual features [3, 102]. In manipulation, generated frames or subgoal images have been converted into actions [8, 17, 96]. Studies of WAMs compare how the video and action streams should exchange information [75, 86] and how long and how compressed their context should be [31, 47]. Further forecast targets include motion-weighted video [33] and optical flow or motion vectors [15, 42].

3D and motion representations. Object-centric point clouds improve generalization [105], and 3D and 4D VLAs bring 3D and spatiotemporal inputs to large models [93, 99].

Dense pixel motion has also been predicted as an intermediate representation [54, 57, 58]. Beyond SE(3) decompositions, 3D dynamic scene memories [83] and scene flow estimation [45, 69] model how scenes move. Point trackers, pose estimators, and feed-forward reconstruction models [34, 73, 76, 80, 92] extract such motion labels from video; our DROID pretraining uses the 3D point tracks that PointWorld extracted [30] (Appendix A.5).